(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
07 December 2023 (07.12.2023)

(10) International Publication Number
WO 2023/233187 A1

(51) International Patent Classification:
G01N 33/24 (2006.01) *A01C 21/00* (2006.01)
G01N 23/22 (2018.01) *A01B 39/00* (2006.01)

(21) International Application Number: PCT/IB2022/055177

(22) International Filing Date: 03 June 2022 (03.06.2022)

(25) Filing Language: English

(26) Publication Language: English

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(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(54) **Title:** PORTABLE DEVICE AND SOIL ANALYSIS METHOD AND ITS CALIBRATION TO CULTIVATING APPROPRIATE PLANTS

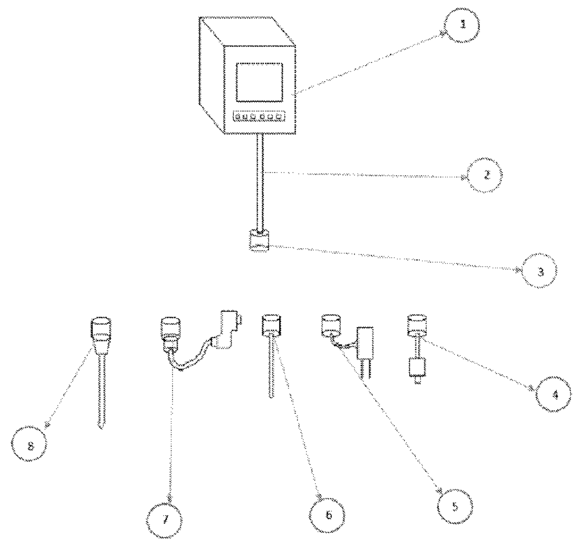

Fig 1

(57) **Abstract:** Abstract In this device, the central digital system acts as a center for analysis and storage and transferring the measured data to the application, and several testing tools are designed appropriately for this device as different series. Upon performing the desired tests and storing the data related to each of the tests in the memory of the digital part of the device, such data are analyzed and evaluated, and the properties of the soil are determined in a specific way. Utilization of these data requires a database and knowledge of botany, as due to the existence of many different species of plants, learning, memorization, and even remembering them is problematic. For this, a specialized application that contains traits of plants and trees can assess the appropriate location and suitable properties for planting and growing plants and trees in a categorized manner.

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[Continued on next page]

(84) Designated States *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SC, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:

— *with international search report (Art. 21(3))*

— *in black and white; the international application as filed contained color or greyscale and is available for download from PATENTSCOPE*

Description

Title of Invention : Portable device and soil analysis method and its calibration to cultivating appropriate plants

Technical Field

[0001] The technical field of this invention relates to devices and software for improving the agricultural process at home and industrial scale.

Background Art

[0002] AGRICULTURAL DEVICES, SYSTEMS, AND METHODS OF DETERMINING CHARACTERISTICS OF SOIL AND SEEDS AND THEIR ANALYSIS filed in the Russian Federation. the agricultural system of sowing seeds in all three embodiments comprises a data processing unit, a frame, and an opener connected to the frame for cutting furrows in the soil. In addition, the system also comprises a sensor connected to the data processing unit and configured to receive a characteristic connected with sowing seeds. At that, the sensor generates a signal related to the perceived characteristic, and the data processing unit receives this signal.

[0003] And the said signal is associated with sowing seeds, and the processing unit uses this signal before sowing the next seed. And the said sensor is selected from the group consisting of an image sensor in the visible region of the spectrum, an ultrasonic sensor, a capacitive sensor, a photoelectric sensor, a luminescence sensor, a contrast sensor, a video camera, a color sensor, and a laser distance sensor. The second embodiment differs from the first in that the sensor is used, which perceives such characteristics associated with sowing as soil moisture. The third embodiment differs from the first two embodiments in the presence of the second sensor made with the ability to determine a second characteristic related to sowing.

[0004] While this system is very similar to the claimed invention, in the claimed invention various parameters to reach a better and more accurate result are evaluated.

Technical Problem

[0005] To obtain an appropriate crop yield, plant cultivation entails for various conditions. One of the most determinant factors is an assessment of soil properties in a given area so that soil should have desirable conditions and enormous factors such as metals concentration in the soil, moisture and acidic and alkaline conditions and toxic substances in the soil should be taken into account.

[0006] Another issue is the adaptation of plant conditions to soil quality, and one of the permanent issues and challenges is to assess which plants are suitable for soil conditions and what percentage of success is expected. At the same time, the requirement for high expertise to assess the soil properties and adaptation to plant conditions incurs huge costs as well as time or in other words trial and error in this area.

[0007] Given these basic issues, as well as other cases such as on-site investigations and so on, a multi-purpose portable device with proper accuracy and a suitable application has been designed.

Solution to Problem

[0008] As was discussed before, given the present obstacles and issues, as well as the huge cost and time-consuming tests, this device is designed in such a way that it obviates many of these issues. In this device, the central digital system acts as a center for analysis and storage and transferring the measured data to the application, and several testing tools are designed appropriately for this device as different series.

[0009] The central digital device is equipped with a CPU for data processing and analysis, the corresponding operating system, a Wi-Fi system for Internet connection and application supplied by a SIM card, a hard disk and RAM for data storage, a battery for power supply, USP sockets for using the USP cable to transmit data manually as well as charging batteries.

[0010] Upon performing the desired tests and storing the data related to each of the tests in the memory of the digital part of the device, such data are analyzed and evaluated, and the properties of the soil are determined in a specific way.

[0011] Utilization of these data requires a database and knowledge of botany, as due to the existence of many different species of plants, learning, memorization, and even remembering them is problematic. For this, a specialized application that contains traits of plants and trees can assess the appropriate location and suitable properties for planting and growing plants and trees in a categorized manner.

[0012] It can be installed on mobile phones and all kinds of computers, the data can be transferred to it, and then this software offers the desired plant options according to our demands. Among the features and capabilities of this application are the use of product diversity and filtering of individual options by goals.

[0013] This device assesses soil according to the special testing method for each series and analyzes the suitable data according to the features of this device, the following sampling series is designed and embedded for this invention:

[0014] • Soil moisture meter

[0015] The soil moisture level of the site is measured using special moisture test series, which is designed as a digital rod hygrometer, data are analyzed in the digital part of the device, and then these data are stored.

[0016] • Soil chemical analysis device

[0017] The special series for chemical analysis and the testing device is equipped with a digital spectrometer that uses UV spectroscopy to assess and test chemicals and the microbes' levels in the soil. Determining chemicals in the soil in addition to suitable plant selection, analysis the health of the soil and can improve soil properties and strength.

[0018] • Soil acidity and salinity test machine

[0019] A special series for acidity and salinity testing in the form of two metal electrodes that are connected to the digital part of the device through a cable and has the potential to conduct electrical current and to measure soil acidity and salinity as very determinant parameters

[0020] • Plasma device for soil metals analysis

[0021] The concentration and type of metals in the soil are analyzed by its special series in a device equipped with an X-ray reflecting tube in the soil texture and spectroscopic results are obtained differently from laboratory cases. The device is portable and provides a good estimate on the site in an accurate way.

[0022] • Soil Hydrometric device for measuring soil density

[0023] Soil density can be measured by special series equipped with a heating tube, pycnometer, and weighing system. As it can be seen, nearly all specialized and important soil tests in this device are conducted simultaneously so that the device is designed and made as a multi-purpose invention.

[0024] Finally, for instance, to assess the best recommendation for soil with the following conditions, the following results can be reached:

[0025] • Soil pH : 6

[0026] • Elements and metals in soil: aluminum sulfate, iron sulfate, nitrogen

[0027] • Soil toxin: formaldehyde, xylene, and toluene

[0028] • Soil moisture: 30%

[0029] • Soil density and texture: North light porous soil consisted of 50% sand, perlite, and 50% leaf mold and peat moss and clay

[0030] Through evaluating the samples in the list, the best plant for this soil is Censoria, which under suitable light conditions will have a very high establishment rate.

Advantageous Effects of Invention

[0031] The most important advantages of using this device in comparison with similar products are as follow:

[0032] • Incorporation of multiple products in one multi-purpose device

[0033] • Cost-effectiveness of soil tests

[0034] • Desirable accuracy to get initial estimates

[0035] • Improving decision making in plant selection

[0036] • Short time of analysis and tests

[0037] • Portability and obviation of need for costly transportation

[0038] • It can be used in a variety of mobile phones

[0039] • Data can be transferred to specific analysis software

[0040] • It is very cost-effective compared to similar testing devices

[0041] • It provides many choices according to the traits of plants related to soil type

[0042] • It is equipped with a specialized application as a specialist and consultant

Brief Description of Drawings

[0043] Figure 1: General components of the device

[0044] Figure 2: Components of Central digital device

[0045] Figure 3: Components of Soil chemical testing device

[0046] Figure 4: Components of a hydrometric device for testing soil density

[0047] Figure 5: Components of soil moisture meter

Description of Embodiments

[0048] The body (1-1) of the central digital device (1) consists of PVC with high impact resistance. A 5-inch display (2-1) is installed on a central digital device that displays the essential information. Analog buttons (1-3) have been used to devise manual settings.

[0049] USB port (1-4) is embedded to connect to the cable for manual charging and data transfer. The hardware features of the central digital device are mounted on the electronic and digital board (1-5). To store and keep the data being processed the RAM of the central digital device (1-6) is used.

[0050] The central digital device system is equipped with Wi-Fi (1-7). Hard disk (1-8) is used to store data. CPU (1-9) is used to process and analyze the data and the corresponding operating system. The SIM card (1-10) is considered to use the SIM card Internet and send SMS. Connection cable (2) and socket (3) are used to connect the central digital device to the device series.

[0051] A soil chemical testing device (4) to analyze and test the soil chemicals has been used, which is equipped with a digital spectrometer and has been designed, calibrated, and made using UV spectroscopy. This device is connected to the central device by socket (4-1) and cable (4-2).

[0052] UV irradiation and transmission system (3-4) is considered for digital spectrometry to UV spectrum irradiation and chemical analysis of substances in the soil. Soil acidity and salinity testing device (5) is utilized to test acidity and salinity. This device consists of two metal electrodes with the potential to conduct an electric current and measure the soil acidity. It is also able to measure the soil salinity. The information received is analyzed and processed by the digital part of the soil acidity and salinity testing device (5-1). The soil acidity and salinity testing device consist of two metal electrodes (2-5) with the potential to conduct an electric current to measure the acidity and salinity. This device is connected to the main device by cable (5-3) and socket (5-4).

[0053] A hydrometric device (6) is used to measure soil density. This device is embedded with a socket (6-1) to connect to the main device. This device is equipped with a heating tube (6-2) and a pycnometer and a weighing system for soil density.

[0054] A plasma device (7) equipped with a reflection X-ray tube is used to analyze the level and type of metals in the soil texture. This device is designed, calibrated, and made using the XRF system and receives the spectroscopy results inappropriate to the size of this device. The digital part of the device (7-1) performs X-ray irradiation using an XRF system in the soil and analysis of soil metals. Cable (7-2) and socket (7-3) are used to connect with the digital part of the plasma device for metals analysis in the soil.

[0055] The soil moisture meter (8) operates using a digital rod hygrometer (8-2) which is designed, calibrated, and made g the size of this device. Socket (8-1) is used to connect this device to the central digital device.

Examples

[0056] This device is implemented as follows: first, considering the depth of the soil specific series (considering the given depth, soil can be drilled and sampled at that site), the number of soil samples are extracted and tested, and special series are entered. Subsequently using the digital system connected to it, the necessary data are extracted and analyzed and finally are transferred into the special application installed on the phone. The results of the analysis are displayed through a search in the application database for product selection in the form of a list.

Industrial Applicability

[0057] This device is designed for soil analysis, so it can be used in any site that requires soil analysis for plants or it can be utilized in cases the parameters can be analyzed by the device, for example, both for agricultural lands and analysis of plant cultivation in home gardens and pots, as well as to analyze specific soil tests.

Claims

[Claim 1] A device and soil analysis method for cultivating appropriate plants comprising:

a. Soil moisture meter in a way that the soil moisture level of the site is measured using special moisture test series, which is designed as a digital rod hygrometer, data are analyzed in the digital part of the device, and then these data are stored;

b. Soil chemical analysis device in a way that the special series for chemical analysis and the testing device is equipped with a digital spectrometer that uses UV spectroscopy to assess and test chemicals and the microbes' levels in the soil;

c. Soil acidity and salinity test machine in a way that a special series for acidity and salinity testing in the form of two metal electrodes that are connected to the digital part of the device through a cable and has the potential to conduct electrical current;

d. Plasma device for soil metals analysis in a way that the concentration and type of metals in the soil are analyzed by its special series in a device equipped with an X-ray reflecting tube in the soil texture and spectroscopic results are obtained differently from laboratory cases;

e. Soil Hydrometric device for measuring soil density in a way that the soil density can be measured by special series equipped with a heating tube, pycnometer, and weighing system;

a database and knowledge of botany, that contains traits of plants and trees will assess the appropriate location and suitable properties for planting and growing plants and trees in a categorized manner.

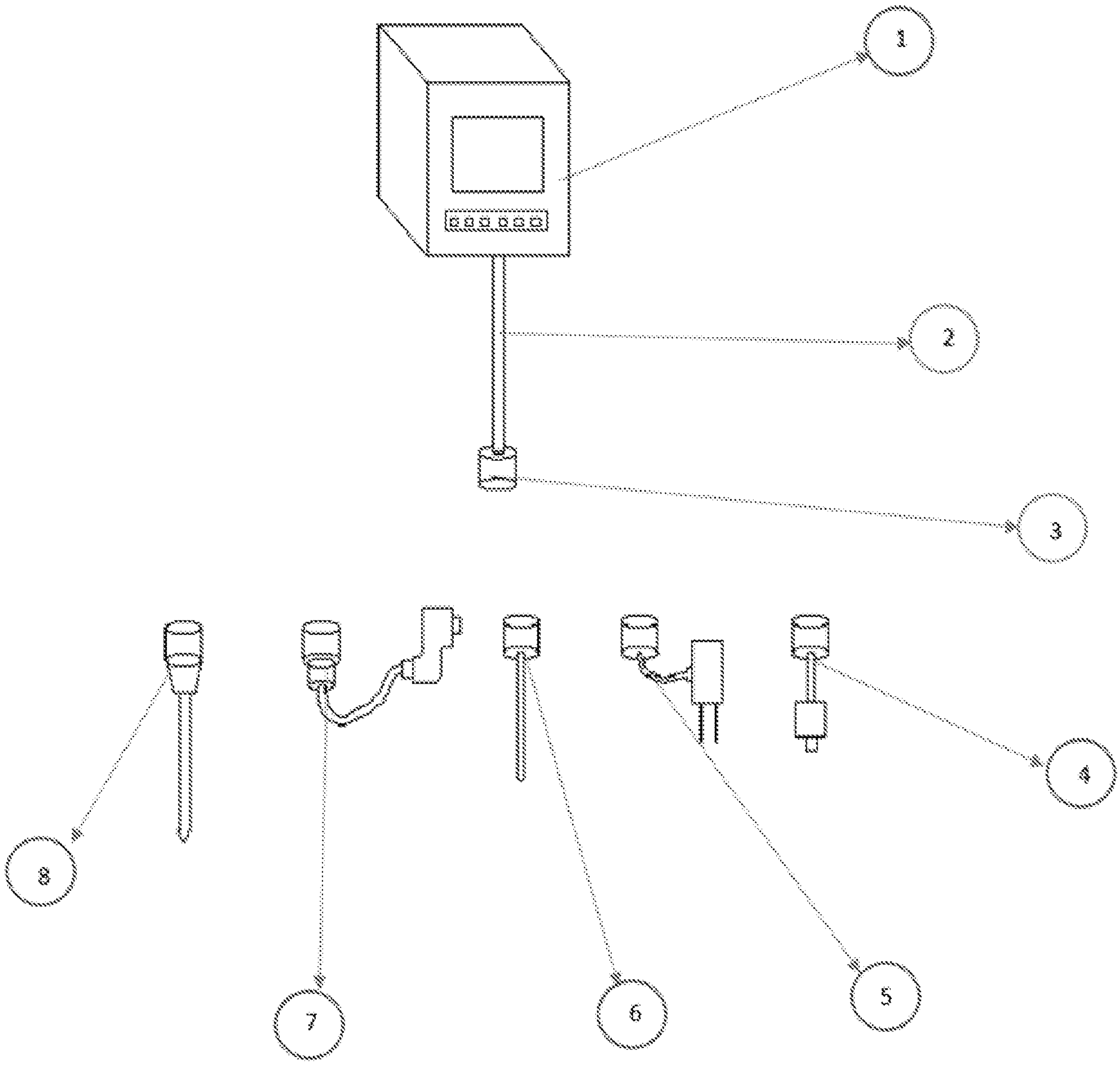

Fig 1

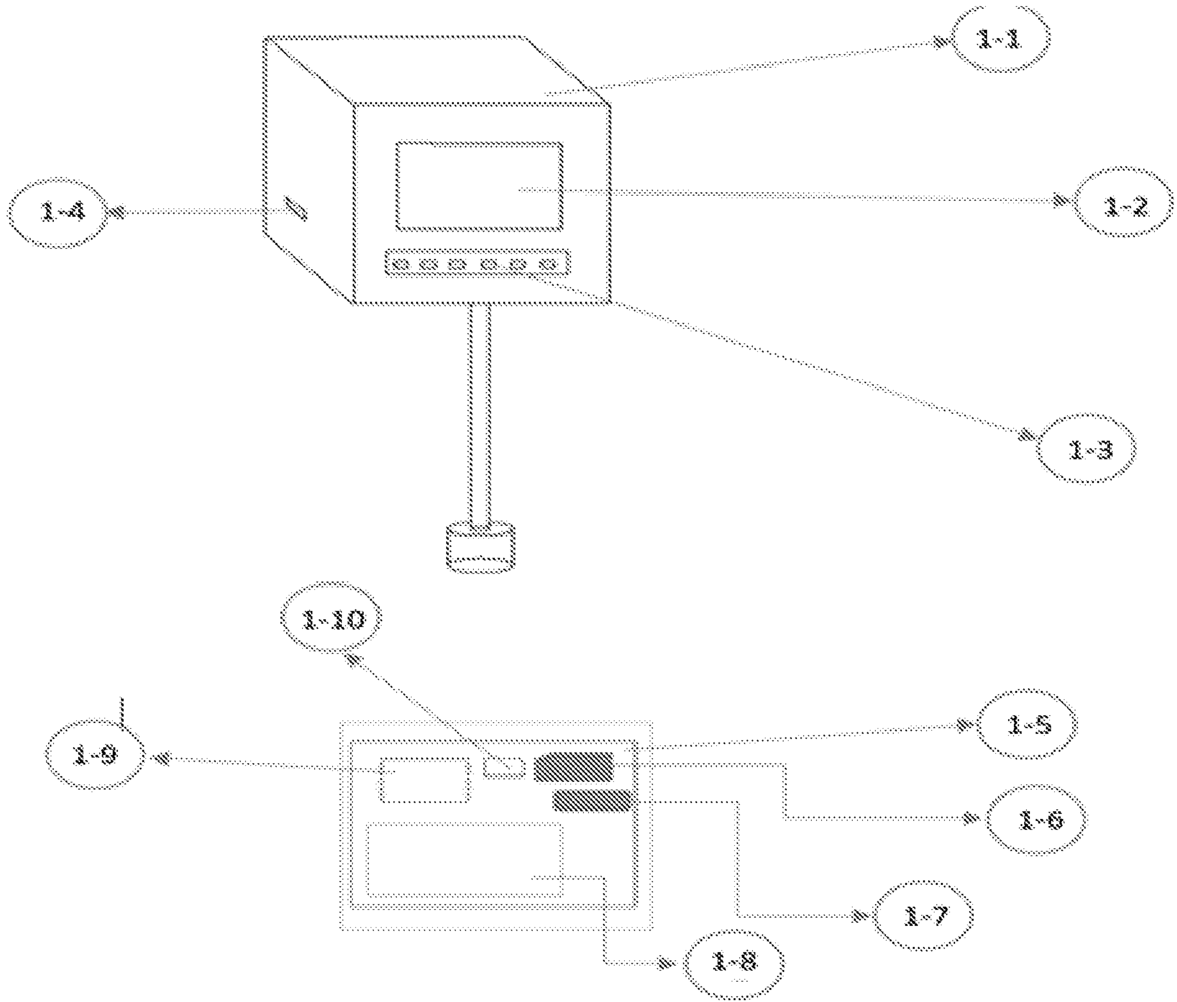

Fig 2

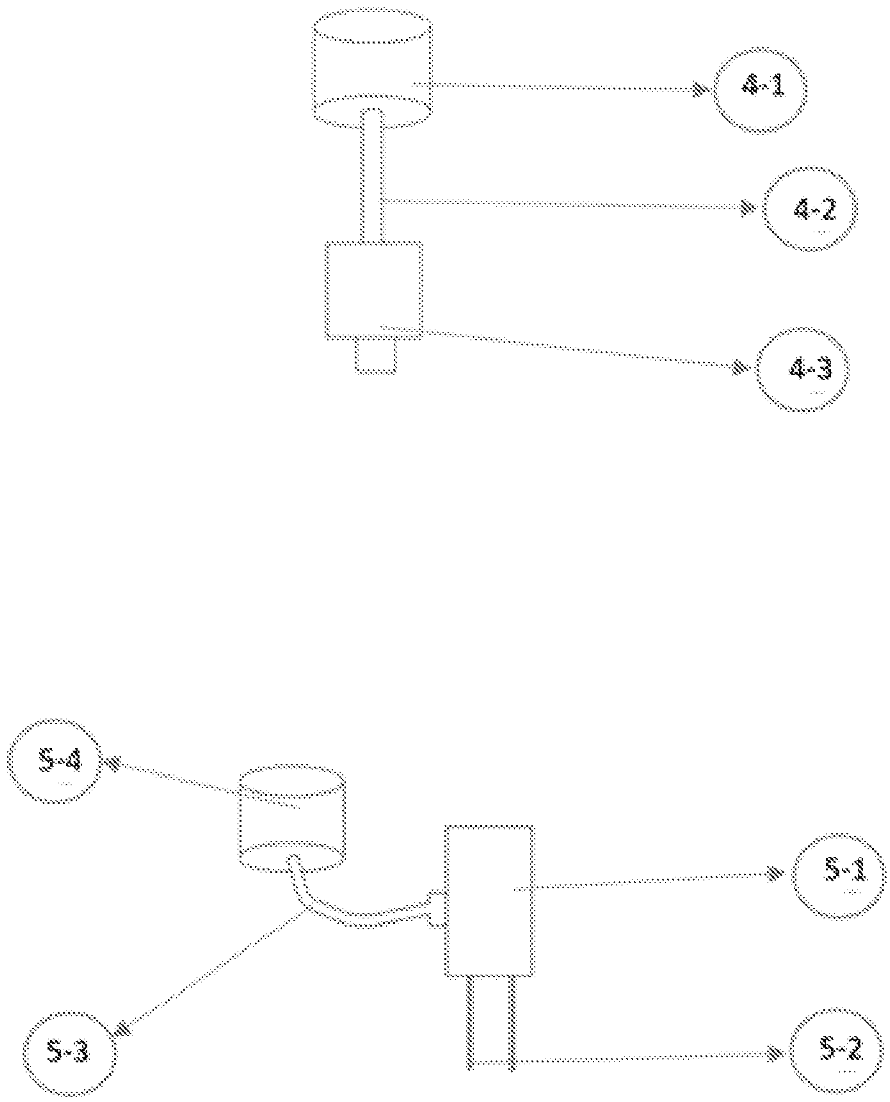

Fig 3

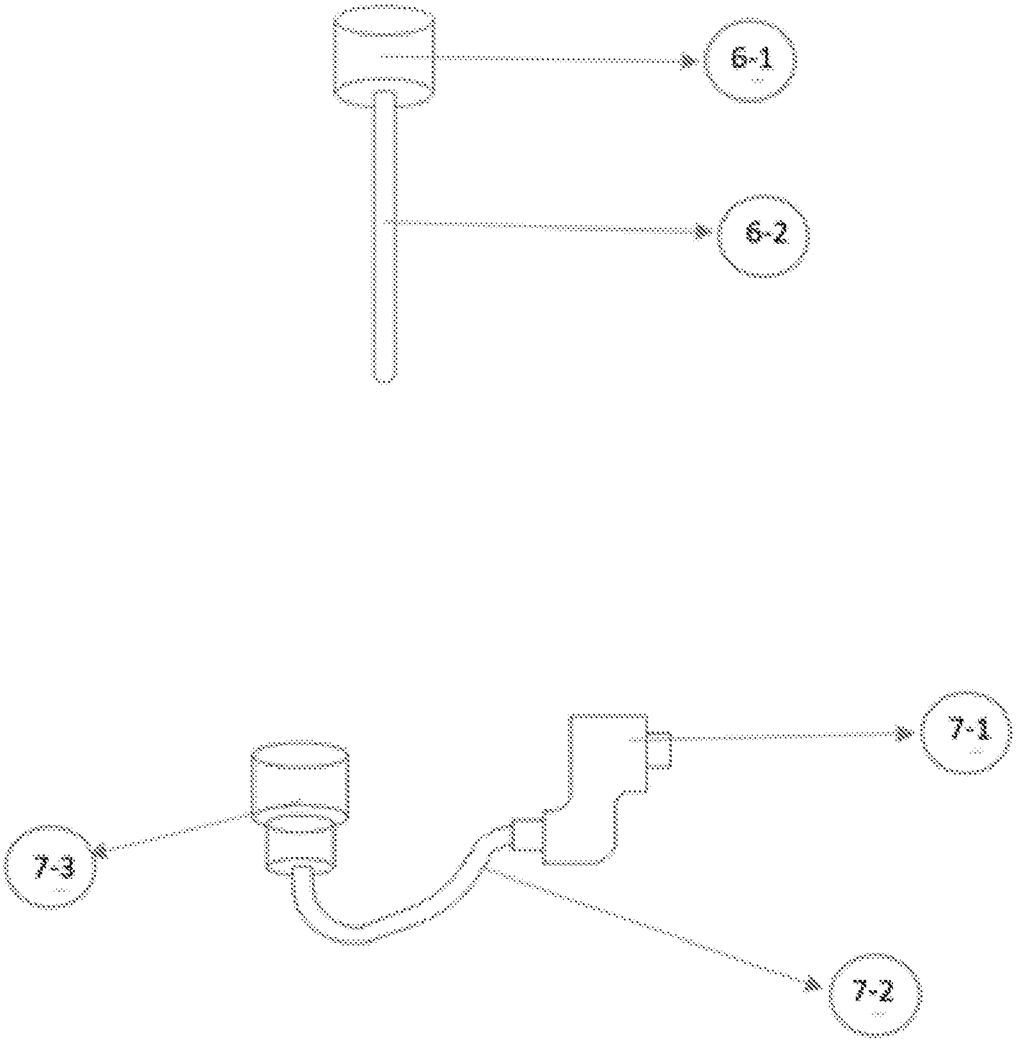

Fig 4

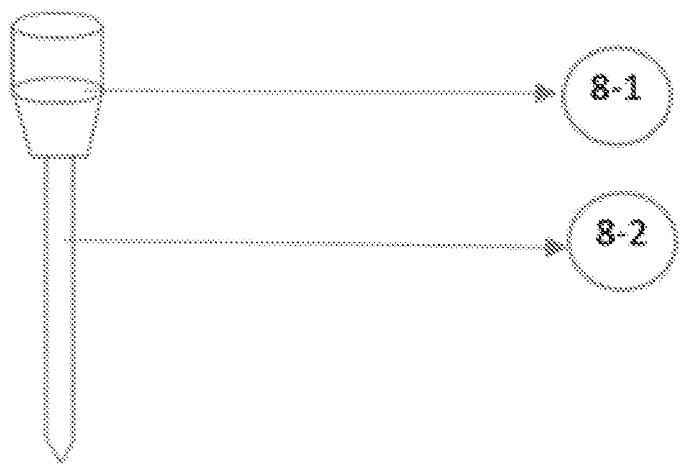

Fig 5

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2022/055177

A. CLASSIFICATION OF SUBJECT MATTER

G01N33/24,G01N23/22,A01C21/00,A01B39/00 Version=2022.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01N, A01C, A01B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

Databases: Patseer, IPO Internal Database
Keywords: soil, moisture, spectrometer, saline, hydrometer, crop

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	US20170122889A1 [UNIV TEXAS TECH SYSTEM; RAMAKRISHNA MISSION VIVEKANANDA UNIV] 04 May 2017 (04-05-2017) Paragraphs [0019], [0061], [0121]-[0128]; claims 1-17; figures 1-4 & 26.	1
Y	US20140012732A1 [TRIMBLE NAVIGATION LTD] 09 January 2014 (09-01-2014) Paragraphs [0158]-[0163]; claims 1-17; figures 1-5 & 6-11.	1
A	US20180188225A1 [COMMW SCIENT IND RES ORG] 05 July 2018 (05-07-2018) Paragraphs [0009]-[0015], [0071]-[0125]; claims 1-19; figures 1, 2, 5, 9, 16-19 & 25.	1

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:
- "A" document defining the general state of the art which is not considered to be of particular relevance
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Date of the actual completion of the international search	Date of mailing of the international search report
26-08-2022	26-08-2022

Name and mailing address of the ISA/	Authorized officer
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Citation	Pub.Date	Family	Pub.Date
US 2017122889 A1	04-05-2017	WO 2015195988 A1	23-12-2015
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